

**Identifying the barriers and facilitators of remote patient monitoring implementation in
hospital-based maternity care: a scoping review**

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Abstract

Background: The United States has the highest maternal mortality rate among high-income nations, with significant disparities in health outcomes. Remote patient monitoring (RPM) leverages digital technology to facilitate continuous management of medical conditions, which allows clinicians to detect early signs of decline and intervene. While RPM has been shown to improve chronic disease management during the perinatal period, there is limited research on the barriers and facilitators of implementation in hospital-based maternity care.

Methods: We reviewed studies published between January 2010 and December 2025 using PubMed, Science Direct, Embase, and Web of Science. Search terms included variants of maternal health, remote patient monitoring, and implementation. The inclusion criteria were: 1) published in a peer-reviewed journal; 2) published between January 2010 to December 2025; 3) located in the United States; 4) published in English; 5) described the implementation of a RPM intervention for maternity care; 6) takes place in a hospital setting; and 7) full-text available. One independent reviewer performed searches, screening, data extraction, and quality assessment. Each included study was then analyzed using the Consolidated Framework for Implementation Research (CFIR).

Results: A total of 889 studies were generated from the initial search, with 152 exclusions due to duplicates. Following screening at the title/abstract and full-text levels, 19 studies were included in the final review. Barriers and facilitators of RPM implementation were identified in each domain of the CFIR. Key facilitators include pre-established hospital digital infrastructure and flexible RPM training for providers, with telehealth policies further streamlining implementation. Conversely, barriers include cost constraints, EHR integration challenges, and provider burnout from high volumes of incoming data.

Conclusion: The study determined that implementation in maternity care presents unique challenges like time-sensitive care and coordination of diverse care teams. To enhance perinatal RPM implementation, hospitals should design RPM workflows with clinicians, invest in data visualization and interpretation tools, expand staff capacity, and ensure devices are designed to accommodate varying levels of digital health literacy.

Keywords: *remote patient monitoring, Consolidated Framework for Implementation Research, maternity care, scoping review*

Background

The United States has the highest maternal mortality rate (MMR) among high-income countries, with 18.7 deaths per 100,000 live births reported in 2023.¹ The most common causes of preventable maternal death are severe bleeding, sepsis, cardiovascular conditions, and behavioral health challenges.¹ Additionally, there are significant racial and ethnic disparities in pregnancy-related deaths. In 2023, the MMR for Non-Hispanic Black people was 49 deaths per 100,000 live births compared to 16 deaths per 100,000 live births for Non-Hispanic White people.¹ Evidently, the U.S. has large disparities in state health system performance, with key contributors being quality, access, affordability, and prevention.² These health disparities are oftentimes associated with poor hospital quality ratings for maternal health.³ Hospitals are rated according to mortality, safety of care, readmission, patient experience, and timely and effective care. Specifically, hospital maternal health is rated using the maternal morbidity structural measure, caesarean birth measure, and severe obstetric complications measure.⁴

One emerging strategy to reduce pregnancy-related deaths and improve hospital perinatal quality is the integration of remote patient monitoring (RPM) into maternity care models.⁴ RPM leverages digital technologies to gather and share health data between patients and clinicians for continuous management of medical conditions.⁵ These tools allow clinicians to detect early signs of clinical decline and intervene early, reducing preventable complications. Studies suggest that leveraging RPM during the perinatal period can reduce emergency visits, improve blood pressure ascertainment, and increase patient satisfaction.⁶ RPM may also be particularly beneficial for patients who face barriers to in-person care, such as lack of transportation, childcare responsibilities, and inflexible work schedules.⁷ Virtual monitoring also reduces opportunities for

biased decision-making by limiting the influence of systemic racism on in-person clinical interactions.⁸

The adoption of RPM and other telehealth platforms has accelerated in recent years due to the COVID-19 pandemic.⁷ Between 2018-2022, there was a 40.3% increase in the number of U.S. hospitals offering RPM services with the goal of monitoring chronic conditions.⁹ Achieving optimal outcomes of RPM services requires not only service design but also effective implementation. Currently, there remains minimal research on the barriers to and facilitators of RPM implementation within maternal health settings in U.S. hospitals. RPM implementation faces challenges like patient engagement, clinician workflows, and electronic health record (EHR) data integration systems.¹⁰ To effectively reform the delivery system and ensure the best obstetric outcomes, it is important that these factors are well understood.

Therefore, the question this scoping review answers is: What are the barriers and facilitators of implementing RPM into maternity care within U.S. hospitals? This was achieved using the Consolidated Framework for Implementation Research (CFIR), which is a determinant framework that includes constructs from multiple implementation theories, models, and frameworks.¹¹ Since the CFIR has been most used to test implementation success amongst interventions in health facilities, we deemed it relevant to our project as it allows us to retrospectively identify the barriers and facilitators of hospital-based RPM implementation. The results inform a set of priorities for future RPM implementation in hospital-based maternity care.

Methods

A scoping review was conducted according to the guidelines presented by the Joanna Briggs Institute (JBI) guidelines.¹² The objectives, inclusion criteria, and methods for this scoping review were specified in advance and documented in a protocol. Additionally, the

120 Preferred Reporting Items for Systematic Reviews and Meta-analyses Extension for Scoping
121 Review (PRISMA-ScR) checklist was used to guide the reporting of methodology and results.¹³

122 **Consolidated Framework for Implementation Research**

123 The CFIR encompasses 48 constructs across 5 broad domains: 1) Innovation; 2) Outer
124 Setting; 3) Inner Setting; 4) Individuals: Roles & Characteristics; and 5) Implementation
125 Process.¹¹ Data gathered from the scoping review is categorized within each of the five domains
126 to understand the various factors that impact implementation outcomes. For this paper, only the
127 domains and constructs deemed relevant to the included studies are assessed. The definitions of
128 each domain of the CFIR, along with their associated constructs that this paper identifies are
129 presented in Appendix 1.

130 **Search strategy**

131 Four electronic databases with a scope in healthcare were used to search for studies
132 between January 2010 through December 2025: PubMed, Science Direct, Embase, and Web of
133 Science. Search terms focused on maternal health (i.e., maternal health, pregnancy care,
134 obstetrics, gynecology, perinatal care), remote patient monitoring (i.e., remote patient
135 monitoring, telehealth), and implementation (i.e., implementation, integration, provision).
136 Finalized searches were conducted during January 2026. All included studies were hand
137 searched to identify additional articles. The full search strategy is available in Appendix 2.

138 **Inclusion Criteria**

139 Studies were selected if they met the following inclusion criteria: (1) published in a peer-
140 reviewed journal; (2) published between January 2010 to December 2025; (3) located in the
141 United States; (4) published in English; (5) described the implementation of a RPM intervention
142 for maternity care; (6) takes place in a hospital setting; and (7) full-text available. We did not

search for gray literature, defined as text not published in manuscripts (e.g., conference abstracts, reports, unpublished trials).¹² Studies were excluded if they: (1) were not based in the United States; (2) did not describe an RPM intervention; (3) focused on RPM implementation in settings outside of maternity care; (4) described an intervention outside the hospital setting; (5) explored the association between RPM and maternal health outcomes, rather than implementation.

Study selection

Studies returned from this search in each database were uploaded into Rayyan, a web and mobile app for systematic reviews.¹⁴ Duplicates were omitted by Rayyan, and the remaining studies were manually screened by one reviewer (AA) at the title and abstract level against the inclusion and exclusion criteria. Finally, the same reviewer (AA) conducted a full text review of eligible studies based on the inclusion criteria.

Data Extraction and Quality Assessment

Data extraction was conducted in Microsoft Excel. Elements extracted from each paper included: title, objective, study design, and study setting. Quality assessment was also performed in Microsoft Excel using the JBI Critical Appraisal Checklist for Expert Opinions or the Mixed Methods Appraisal Tool (MMAT), depending on study design.^{15,16} For each study, we created a quality score to summarize the proportion of items in the assessment that were satisfied.

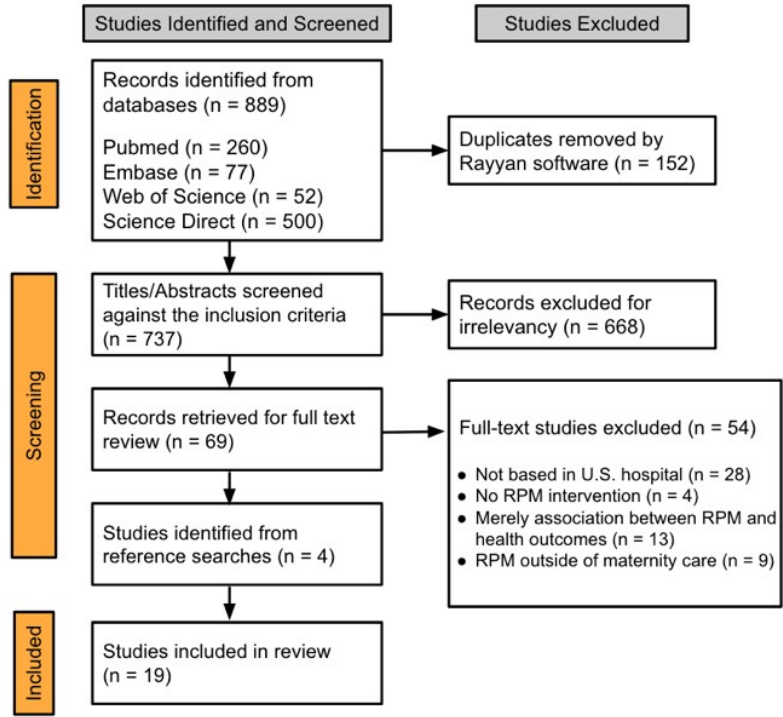
Using the CFIR, we identified the percentage of included studies that discussed each of the five domains in their RPM implementation practices. Then, each domain's relevant constructs were analyzed, and specific barriers and facilitators were summarized. Constructs were identified using the CFIR's domain and construct definitions in Appendix 1.

Results

Included Studies

A total of 889 studies were generated from our search, and 152 duplicates were omitted by Rayyan. The remaining 737 studies were screened at the title and abstract level, which resulted in 668 exclusions due to irrelevancy. The remaining 69 studies were screened at the full-text level for relevance and eligibility. Most studies that were excluded during the full-text screening were either not based in U.S. hospitals (n=28), did not describe an RPM intervention (n=4), merely explored the association between RPM and maternal health outcomes (n=13), or described an RPM intervention in settings outside of maternity care (n=9). After this process, studies were hand-picked from reference searches (n=4). Ultimately, 19 studies were included in the final review (see Figure 1).

Figure 1. PRISMA Flow Chart.



Characteristics of the Included Studies and Quality Analysis

The study characteristics and quality scores for each included study are presented in Table 1. Overall, the included studies met recommended quality criteria 98.5% of the time -- 100% for expert opinions and 98% for qualitative, quantitative, and mixed methods studies. See

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Table 1 for summary quality assessment data. See Appendix 3 and 4 for complete quality assessment tables for qualitative, quantitative, and mixed methods studies, and expert opinions, respectively.

Among the included studies, eight have a mixed-methods study design,^{18, 21-23, 25, 26, 32, 33} six a qualitative study design,^{19, 24, 28, 29, 31, 34} two a quantitative study design,^{20, 27} and three are expert opinions.^{17, 30, 35} Thirteen of the studies are based in urban, academic medical centers,^{17, 18, 21, 22, 25, 26, 28-32, 34, 35} while six studies are either rural, non-profit, or safety net hospitals.^{19, 20, 23, 24, 27, 33}

Table 1. Summary of Included Studies and Quality Assessment

Citation	Title	Objective	Study Design	Location	Quality (%)
Aziz et. al. (2020) ¹⁷	Telehealth for High-Risk Pregnancies in the Setting of the COVID-19 Pandemic	Review the adoption of telehealth for high-risk obstetric patients receiving care at two hospitals and an affiliated clinic system in New York City	Expert opinion; Qualitative data	Large, urban academic medical center in New York City, NY	100%
Delgado et. al. (2023) ¹⁸	Evaluation of Virtual Prenatal Care for Obstetric Care Delivery During the COVID-19 Pandemic: A Mixed Method Research Study Using the Consolidated Framework in Implementation Research	Assess and understand the patient perspectives of prenatal telemedicine utilizing the CFIR	Mixed methods; Quantitative and qualitative data	Large, urban academic clinic in Tampa, FL	100%
Dullabh et. al. (2023) ¹⁹	Implementation and evaluation of an electronic health record-integrated app for postpartum monitoring of hypertensive disorders of pregnancy using patient-contributed data collection	Evaluate a pilot intervention of a smartphone application for postpartum monitoring of hypertensive disorders of pregnancy.	Qualitative evaluation	Large, urban safety net hospital in New Haven, CT	100%
Forna et. al. (2024) ²⁰	Improving obstetric and perinatal outcomes with a remote patient monitoring program for hypertension in a large integrated care system	Determine the effect of a remote patient monitoring program for hypertensive disorders of pregnancy.	Matched retrospective cohort design; Quantitative data	Large, urban non-profit integrated healthcare system in Atlanta, GA	100%

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Hacker et. al. (2022) ²¹	Implementation of a universal postpartum blood pressure monitoring program: feasibility and outcomes	Explore the feasibility of a patient-driven postpartum home blood pressure monitoring program with hypertensive disorders of pregnancy	Prospective observational study; Quantitative and qualitative data	Large, urban multispecialty academic medical center in Pittsburgh, PA	100%
Hofmann et. al. (2022) ²²	Virtual Prenatal and Postpartum Care Acceptability Among Maternity Care Providers	Understand maternity care providers' acceptance of and barriers to providing virtual maternity care.	Descriptive study; Quantitative and qualitative data	Large, urban academic medical center in Aurora, CO	86%
Howard et. al. (2025) ²³	Evaluation of a remote blood pressure monitoring program during pregnancy: utilization and implementation outcomes	Identify factors associated with enrollment, retention, and patient engagement in the antenatal remote blood pressure monitoring program, Connected MOM, and evaluate program implementation by race and residence	Retrospective cohort study; Quantitative and qualitative data	Non-profit, academic healthcare system in rural Louisiana	100%
James et. al. (2025) ²⁴	Obstetricians' experiences with remote monitoring programs for hypertensive disorders	Explore obstetricians' experiences with RPM for hypertensive disorders in the perinatal period and barriers in implementation	Qualitative study	Hospital-based outpatient clinics in the South, Midwest, Northeast, and West of the U.S.	100%
Janssen et. al. (2021) ²⁵	Implementation of a text-based postpartum blood pressure monitoring program at 3 different academic sites	Measure the program participant response rates during the first 10 days after hospital discharge, to assess the replicability and scalability of the program, and to evaluate patient satisfaction at 3 distinct sites	Prospective observational cohort study; Quantitative and qualitative data	Large, urban medical centers in Philadelphia, PA, Chapel Hill, NC, and Chicago, IL	100%
Jeganathan et. al. (2020) ²⁶	Adherence and acceptability of telehealth appointments for high-risk obstetrical patients during the coronavirus disease 2019 pandemic	Describe patient and provider attitudes toward telehealth for the delivery of high-risk obstetrical care and to determine whether the implementation of a telehealth model improves patient adherence to scheduled appointments	Cross-sectional descriptive survey; Quantitative and qualitative data	Large healthcare system with both urban and suburban settings in New York, NY	100%
Klein et. al. (2024) ²⁷	The feasibility of implementing a digital pregnancy and postpartum support program in the	Assessed associations between the Pregnancy Postpartum Support Program (PPSP), a digital	Retrospective analysis; Quantitative data	Integrated non-profit healthcare system in rural Illinois	100%

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	Midwestern United States and the association with maternal and infant health	wraparound service, and maternal and infant outcomes in a Medicaid population			
Lawrence et. al. (2023) ²⁸	Operational Implementation of Remote Patient Monitoring Within a Large Ambulatory Health System: Multimethod Qualitative Case Study	Evaluate the early implementation of operational RPM initiatives for chronic disease management within the ambulatory network of an academic medical center, focusing on the experiences of “early adopter” clinicians and patients	Multimethod qualitative approach; Qualitative data	Large, urban academic medical center in New York, NY	100%
MacEwan et. al. (2024) ²⁹	Patient and Provider Perspectives About the Use of Patient-Generated Health Data During Pregnancy: Qualitative Exploratory Study	Explore the perspectives of patients and providers about the use of Patient Generated Health Data during pregnancy, including the benefits and challenges of reporting, collecting, and using these data.	Exploratory qualitative study	Large, urban academic medical center in Columbus, OH	100%
Peahl et. al. (2020) ³⁰	Prenatal care redesign: creating flexible maternity care models through virtual care	Explore how insights from RPM implementation can inform patient-centered prenatal care redesign during and beyond the coronavirus disease pandemic	Expert opinion; Qualitative data	Large, urban academic medical center in Ann Arbor, MI	100%
Rajpurkar et. al. (2025) ³¹	Stakeholder Perspectives of New Tailored Prenatal Care Delivery	Evaluate obstetric stakeholders' attitudes and implementation considerations regarding the adoption of key components of a newly tailored prenatal care recommendation	Qualitative study	10 obstetric care departments across the U.S. in rural and urban settings	100%
Reisinger-Kindle et. al. (2021) ³²	Evaluation of rapid telehealth implementation for prenatal and postpartum care visits during the COVID-19 pandemic in an academic clinic in Springfield, Massachusetts, United States of America	Evaluate telehealth implementation in a large academic urban obstetric practice that serves a medically underserved population.	Retrospective cohort study; Quantitative and qualitative data	Large, urban academic medical center in Springfield, MA	100%

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Rhoads et. al. (2017) ³³	Exploring implementation of m-Health monitoring in postpartum women with hypertension	Identify and examine the potential factors that influenced use of m-health technology and adherence to monitoring symptoms related to preeclampsia in postpartum women	Randomized control trial; Quantitative and qualitative data	Rural, academic medical center in Little Rock, AK	86%
Sterba et. al. (2023) ³⁴	Implementation of a women's reproductive behavioral health telemedicine program: a qualitative study of barriers and facilitators in obstetric and pediatric clinics	Characterize the barriers and facilitators to implementing a telemedicine program for the treatment of perinatal mood and anxiety disorders and substance use disorders in obstetric and pediatric providers in the outpatient community practice setting	Qualitative study	Large, urban academic medical center in Charleston, SC	100%
Ukoha et. al. (2021) ³⁵	Ensuring Equitable Implementation of Telemedicine in Perinatal Care	Address implementation factors that should be considered to ensure that disparities are not widened as telemedicine becomes more integrated into perinatal care delivery	Expert opinion; Qualitative data	Large, urban academic medical center in San Francisco, CA	100%
Total Quality Score					98.5%

189

190 Data Extraction Using CFIR

191 Each included study was analyzed using the five CFIR domains and its relevant
 192 constructs. Overall, 100% of the studies describe “Innovation,” 63% the “Outer Setting,” 84%
 193 the “Inner Setting,” 95% the “Characteristics of Individuals,” and 79% the “Implementation
 194 Process.” See Table 2 for the complete analysis of studies that address each domain.

195 **Table 2. Percentage of CFIR domains that were addressed in the studies**

Included Studies	CFIR Domains				
	Innovation	Outer Setting	Inner Setting	Characteristics of Individuals	Implementation Process
Aziz et. al. (2020) ¹⁷	X	X	X	X	X
Delgado et. al. (2023) ¹⁸	X	X	----	X	X
Dullabh et. al. (2023) ¹⁹	X	----	X	X	X
Forna et. al. (2024) ²⁰	X	----	X	X	X
Hacker et. al. (2022) ²¹	X	X	X	X	X

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Hofmann et. al. (2022) ²²	X	X	X	X	X
Howard et. al. (2025) ²³	X	----	X	X	X
James et. al. (2025) ²⁴	X	X	X	X	X
Janssen et. al. (2021) ²⁵	X	----	----	----	----
Jeganathan et. al. (2020) ²⁶	X	X	X	X	----
Klein et. al. (2024) ²⁷	X	----	----	X	X
Lawrence et. al. (2023) ²⁸	X	X	X	X	X
MacEwan et. al. (2024) ²⁹	X	X	X	X	----
Peahl et. al. (2020) ³⁰	X	X	X	X	X
Rajpurkar et. al. (2025) ³¹	X	X	X	X	X
Reisinger-Kindle et. al. (2021) ³²	X	X	X	X	X
Rhoads et. al. (2017) ³³	X	----	X	X	X
Sterba et. al. (2023) ³⁴	X	----	X	X	X
Ukoha et. al. (2021) ³⁵	X	X	X	X	----
Studies, n (%)	19/19 (100%)	12/19 (63%)	16/19 (84%)	18/19 (95%)	15/19 (79%)

196

197 Domain 1: Innovation

198 The Innovation domain describes the product, idea, or program being implemented.¹¹ All
199 19 studies described Innovation, with multiple barriers and facilitators being identified across
200 four constructs. The *relative advantage* of RPM over in-person care was described in 13 studies,
201 with key facilitators being reduced travel time, flexible scheduling, and continuous monitoring.^{17,}
202 ^{18, 20, 22, 26-29, 30, 31, 33-35} However, five studies noted that some patients and providers perceived RPM
203 as the inferior option to in-person care for high-risk obstetric cases due to the absence of life-
204 saving physical exams.^{17, 26, 30-32} Additionally, *innovation complexity* was a recurring barrier in
205 five studies due to multi-step enrollment processes, device pairing difficulties, and provider
206 burnout from alert systems.^{17, 24, 25, 29, 33} *Innovation cost* was also described as a major barrier in
207 five studies for individuals who were not insured, since out of pocket expenses
208 disproportionately affected low-income communities.^{20, 24, 29, 34, 35} Furthermore, from the provider
209 perspective, transitioning from nonautomated to automated RPM was too expensive, due to

equipment, software, and insurance costs.²⁴ However, four studies identified that devices that were either given as loaners or covered by insurance expanded the reach of RPM.^{20, 23, 31, 35} Lastly, *innovation design* was described in four studies, with key barriers being poor RPM user interface, insufficient instructions, and interfaces that did not account for health literacy disparities.^{19, 21, 27, 35} On the other hand, RPM devices that had automated response messaging systems and comfortable wear improved adherence, which were both identified as facilitators for implementation.^{17, 19, 24, 33}

Domain 2: Outer Setting

The Outer Setting domain captures macro-level factors (e.g., policy, economic activity, etc.) and how they impact implementation.¹¹ Among the 12 studies that discussed the Outer Setting, the *local conditions* in which RPM programs were implemented acted as a primary facilitator. The COVID-19 pandemic created a large need for telehealth, especially in obstetrics care, making it easier for hospitals to implement RPM programs.^{17, 18, 21, 22, 24, 26, 28-32, 35} Hospitals relied heavily on telehealth to reduce hospital readmissions and protect patients from hospital-acquired infections.²⁴ Additionally, four papers demonstrated how *policies and laws* facilitated the implementation of RPM, especially between 2020-2022. Since the COVID-19 pandemic, some states have enacted explicit payment parity policies, which reimburse telehealth services at the same rate as in-person care.^{31, 32, 35} Additionally, the Centers for Medicare and Medicaid Services (CMS) broadened access to Medicare telehealth services, especially for high-risk obstetrical care in 2020.²⁶ From a regulatory and operational standpoint, this made it easier to implement RPM. However, James et. al. (2025) found that many private insurance plans do not have high reimbursement rates, possibly disincentivizing hospitals from implementing RPM.

Domain 3: Inner Setting

The Inner Setting domain is the area in which the innovation is implemented, including hospitals, schools, or community centers.¹¹ Among the 16 studies that described Inner Setting, six studies mapped barriers to *structural characteristics*. Obstetricians and maternity care providers recorded difficulty with integrating RPM data into EHR, managing large volumes of incoming data, pairing RPM devices with hospital blood pressure machines, and receiving adequate information technology (IT) support.^{19, 22, 24, 32, 33} Specifically, MacEwan et. al. (2024) emphasized the difficulties of reviewing every value and data point from the provider perspective, potentially leading to physician burnout. In contrast, Lawrence et. al. (2023) commented on how presenting data in a way that could help understand trends over time (e.g. tables or charts) acts as a facilitator for structural implementation. Key barriers (described in eight studies) described issues with RPM implementation *compatibility* in hospitals, as it requires clear care coordination, funding, and on-site support if devices do not work. Many providers already have dense portfolios, so integrating a whole new system requires a brand-new workflow and more staff responsibility.^{29, 31, 32, 34} Additionally, RPM requires a new billing and reimbursement system, further complicating these workflows.²⁸ Three studies described how providers fear legal repercussions with receiving RPM nighttime alerts, as they may not be available to respond immediately.^{19, 24, 33} Despite these barriers, three studies described how hospitals that already had preexisting resources (e.g., established digital infrastructure and integrated referral processes) facilitated the implementation of RPM programs in maternity care.^{21, 26, 35}

Domain 4: Individuals - Roles & Characteristics

The Individuals domain highlights the roles and characteristics of individuals implementing, delivering, and/or receiving the innovation.¹¹ In each of the 18 studies in this

domain, *high-level leaders* facilitated RPM implementation, including obstetricians, maternity care providers, advanced nurse practitioners, midwives, and resident physicians.^{17-24, 26-35} Implementation was primarily facilitated in 12 studies due to the *capability* of innovation deliverers (i.e., providers) and recipients (i.e., patients). Providers were generally well-trained through in-person instruction, virtual sessions, emailed guidelines, and ongoing technical support.^{17, 19-21, 26, 30, 32, 34, 35} Patients were also well-trained to use RPM technologies through structured telehealth follow-up to reinforce education.^{20, 27, 33} However, in five studies, capability barriers for patients were primarily due to digital health literacy gaps and cost constraints.^{24, 31, 35} For patients of lower socioeconomic status, it was especially difficult to receive financial and clinical support to best use these technologies.^{19, 22}

Domain 5: Implementation Process

The Process domain documents the activities and strategies being used to implement the innovation.¹¹ All 15 studies in this domain had done adequate *planning* to facilitate implementation. This includes creating workflows with outlined diagnosis, triage, treatment, and disease management to ensure RPM activities are being delivered at each step of the process.^{17-24, 27, 28, 30-34} Additionally, to facilitate *engagement* in RPM, providers would call, text, and email patients to provide education or refer them to other services.^{27, 30} However, a barrier to sustained engagement for patients included privacy and security concerns of technologies.^{24, 33} In two studies, barriers to engagement were heightened for patients in rural regions because of broadband access limitations.^{23, 24} Lastly, *motivation* facilitated ease of RPM implementation in two studies. Both patients and providers valued RPM due to its convenience, however, it also diminished interpersonal connection serving as a barrier to the implementation process.^{18, 28} See Table 3 for a detailed breakdown of barriers and facilitators according to construct and domain.

279 **Table 3. Overview of CFIR domains and constructs that were addressed in the studies as**
 280 **barriers to or facilitators of RPM implementation in hospital-based maternity care**

CFIR constructs by domain	Studies, n (%)	Barriers	Studies, n (%)	Facilitators
Innovation				
Relative advantage	13/19 (68%)	RPM replaces lifesaving in-person physical exams	5/19 (26%)	- Reduced travel time - Flexible schedules for patients - Continuous monitoring of health conditions
Complexity	5/19 (26%)	- Multi-step enrollment - Device pairing difficulties - Provider burnout from alert systems	0	
Cost	5/19 (26%)	- Expensive devices if paying out of pocket - Expensive equipment, software, insurance costs	4/19 (21%)	Insurance coverage of RPM
Design	4/19 (21%)	- Poor RPM user interface - Insufficient instructions for usage - Lack of health literacy for patients	4/19 (21%)	- Automated response messaging systems - Comfortable wearable devices
Outer Setting				
Local Conditions	0		12/12 (100%)	COVID-19 created an urgent need for telehealth
Policies and Laws	1/12 (8%)	Decreased insurance reimbursement rates	4/12 (33%)	- RPM reimbursement through the Centers for Medicare and Medicaid Services - Payment parity for telehealth services
Inner Setting				
Structural Characteristics	6/16 (38%)	- Integrating RPM data into EHR - Managing large volumes of incoming data - Pairing RPM devices with hospital machines - Receiving adequate IT support	1/16 (6%)	Presenting data as trends over time (e.g. tables or charts)
Compatibility	8/16 (50%)	- Integration of new workflow - Care coordination - New billing and reimbursement system - Night alerts and fear of legal repercussion	3/16 (19%)	- Preexisting resources - Familiar technological platforms - Established referral processes
Characteristics of Individuals				
High-level leaders	0		18/18 (100%)	- Obstetricians - Maternity care providers - Advanced nurse practitioners - Midwives

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				Resident physicians
Capability	5/18 (28%)	- Digital health literacy gaps for innovation recipients - Cost constraints for innovation recipients - Limited clinical support	12/18 (67%)	- Innovation deliverers being trained through in-person instruction, virtual sessions, emailed guidelines, and ongoing technical support - Innovation recipients being trained through structured telehealth follow-up to reinforce education
Implementation Process				
Planning	0		15/15 (100%)	Creating workflows with outlined diagnosis, triage, treatment, and disease management
Engaging	3/15 (20%)	- Security and privacy concerns of technologies - Rural individuals having limited access to broadband internet	2/15 (13%)	Call, text, and email patients to provide education or refer them to other services
Motivating	2/15 (13%)	Diminished interpersonal connection between provider and patient	2/15 (13%)	Convenient to use RPM

Discussion

This scoping review used the CFIR to identify the barriers and facilitators of RPM implementation in U.S. hospital-based maternity care. Our findings provide important insights into the operational and administrative factors that should be considered to improve RPM implementation for perinatal care. Ultimately, 19 studies were included in the review, spanning various study designs and settings. Across all five CFIR domains, findings indicated that RPM implementation depends on the interplay between technology, organizational readiness, policy, and individual capabilities.

Main Findings

A facilitator of RPM implementation was its *relative advantage* over in-person care due to reduced travel and continuous monitoring of chronic disease.^{17, 18, 20} However, patients and providers felt that RPM could not fully substitute in-person care for high-risk obstetric cases.^{17, 26, 30-32} Cost and complexity of devices were also recurring barriers, since out-of-pocket RPM expenses fall disproportionately on uninsured and low-income patients.^{20, 24, 29, 34, 35} Programs that

provided loaner or insured devices were more successful in expanding reach.^{20, 23, 31, 35} The COVID-19 pandemic was a common facilitator of RPM implementation across most of the included studies. There were federal expansions of Medicare telehealth coverage and state-level payment parity between 2020 and 2022, which allowed hospitals to implement and get reimbursed for RPM services.²⁶ However, this review found that private insurance reimbursement for RPM services remains inconsistent, and pandemic regulations are beginning to rollback. At the hospital level, EHR integration, high volumes of incoming RPM data, and new workflows created significant barriers in implementation and exacerbated provider burnout.^{19, 22, 24, 32, 33} On the other hand, hospitals with pre-established digital infrastructure and clear referral pathways were more easily able to implement RPM.^{21, 26, 28, 35} Lastly, flexible RPM training for both patients and providers was essential to facilitate implementation. However, these trainings were not equitable as digital health literacy gaps served as a primary barrier for low-income patients to effectively use these technologies.^{21, 35}

Our findings are similar to those of recent reviews examining the implementation of telemedicine and RPM interventions. For example, Hailu et. al. (2024) found that key barriers included the lack of digital health literacy among patients, as well as the inconsistent handling of incoming data by providers.³⁶ Two other studies found that device interface and cost were common barriers, and developing workflows with care teams was critical for success.^{37, 38}

However, this review determined that RPM implementation within maternity care presents unique challenges that sets it apart from other clinical settings. One such challenge is the need for time-sensitive obstetrics care. For example, if a patient is using RPM for preeclampsia management, and a provider has a delayed response to an abnormally high blood pressure alert, the patient may be in a life-threatening situation. Additionally, maternity care involves multiple

overlapping care teams, like obstetricians, midwives, nurse practitioners, doulas, and family physicians. This may complicate workflows and require detailed planning for care coordination. These features mean that RPM priorities designed for maternity care may not be adaptable to other clinical settings.

Implications

By identifying barriers and facilitators of RPM implementation, care teams can improve implementation outcomes (i.e., adoption and sustainment), enhance perinatal care delivery, and ultimately decrease preventable maternal death in the U.S. This review points to four actionable priorities for hospitals to consider in future RPM implementation within maternity care:

- 1) Clinical workflow design should be a foundational implementation activity, created in collaboration with the perinatal care team (i.e., obstetricians, midwives, nurses, and maternal-fetal medicine specialists). Everyone should have clearly defined roles for RPM coordination, including who oversees alert management, diagnosis, triage, treatment, and check-ins.
- 2) To streamline EHR integration and reduce provider burnout, hospitals should invest in data visualization tools, establish clear protocols for interpreting RPM data, and dedicate IT support throughout the implementation process. Providers could also designate external staff to extract notable measures to use as talking points during an appointment with patients.
- 3) Implementation training for providers should be mandatory and include various modes of learning such as in-person instruction, virtual sessions, emailed guidelines, and ongoing technical support. Training content should cover RPM device operation, data

interpretation, and recognition of key obstetric warning signs such as blood pressure trends and postpartum decline.

- 4) To ensure both patients and providers consistently use RPM devices throughout the perinatal period, devices should be designed to be comfortable, easy to understand, and include automated features to account for varying levels of digital health literacy.

Strengths and Limitations

To our knowledge, this is the first scoping review to explore barriers and facilitators of RPM implementation in hospital-based maternity care. A key strength of our review was the use of a comprehensive implementation research framework to guide data collection, analysis, and synthesis. This ensured rigorous methodology for a strong evidence base.

A limitation of our review was that we focused only on hospital-based maternity care, so we cannot generalize our findings to other clinical settings or organizations. Additionally, 68% of our included studies were based in urban, academic medical centers, which may have overstated the operational and administrative capabilities of implementing RPM. Therefore, the proposed priorities may not apply to rural and under-resourced hospitals. We also excluded non-English studies and gray literature from our search, which may have captured the rapid adoption of RPM interventions in recent years. Therefore, we may have missed potentially relevant studies. Lastly, 63% of our included studies were conducted during or after the COVID-19 pandemic, which could have skewed the ease of the RPM implementation process. Further research should explore the Outer Setting, specifically how insurance policies and telehealth regulations impact perinatal RPM implementation. It is vital that provider payment methods and reimbursement policies are understood to incentivize the delivery of value-based care and sustain RPM interventions.

Conclusion

This scoping review provides a detailed interplay of the barriers and facilitators of RPM implementation in maternity care. Key facilitators include pre-established hospital digital infrastructure and flexible RPM training for providers, with telehealth policies further streamlining implementation. Conversely, barriers include cost constraints, EHR integration challenges, and provider burnout from high volumes of incoming data. Implementation in maternity care presents unique challenges like time-sensitive care and coordination of diverse care teams. To enhance perinatal RPM implementation, hospitals should design RPM workflows with clinicians, invest in data visualization and interpretation tools, expand staff capacity, and ensure devices are designed to accommodate varying levels of digital health literacy. Further research is necessary to ensure the sustainment of RPM interventions by exploring how insurance policies impact value-based obstetric care.

Abbreviations

CFIR: Consolidated Framework for Implementation Research

EHR: Electronic Health Record

IT: Information Technology

JB: Joanna Briggs Institute

MMAT: Mixed Methods Appraisal Tool

MMR: Maternal Mortality Rate

PRISMA-ScR: Preferred Reporting Items for Systematic Reviews and Meta-Analyses

Extensions for Scoping Review

RPM: Remote Patient Monitoring

387 **Declarations**

388 **Ethics approval and consent to participate**

389 Not applicable.

390 **Consent for publication**

391 Not applicable.

392 **Availability of data and materials**

393 The datasets supporting the conclusions of this paper are available from the corresponding author
394 upon request.

395 **Competing interests**

396 The authors declare no competing interests

397 **Funding**

398 Not applicable.

399 **Authors' Contributions**

400 A.A. conducted the scoping review and wrote the manuscript. All authors contributed to the
401 study design, reviewed the manuscript, provided feedback, and approved the final manuscript.

402 **Acknowledgements**

403 Not applicable.

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408 **Supplementary Materials**

409 APPENDIX 1. Defined CFIR domains and their associated constructs¹¹

Domain 1. Innovation

The product, idea or program being implemented.

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Construct name	Construct definition
Complexity	The innovation is complicated, which may be reflected by its scope and steps
Cost	The innovation purchase and operating costs are affordable
Design	The innovation is well designed and packaged
Relative Advantage	The innovation is better than other available innovations or current practice
Domain 2. Outer Setting The external environment in which innovation is being implemented.	
Local Conditions	Economic, political, environmental, political, and/or technological conditions enable the implementation and/or delivery of the innovation
Policies and Laws	Legislation, regulations, professional group guidelines and recommendations, or accreditation standards support implementation and/or delivery of innovation
Domain 3. Inner Setting Specific unit where innovation is being implemented.	
Compatibility	The innovation fits with workflows, systems, and processes
Structural Characteristics	Infrastructure components support functional performance of the Inner Setting
Domain 4. Characteristics of Individuals <i>Roles:</i> Extent to which leaders and key actors are involved in implementation efforts and where they sit. <i>Characteristics:</i> The implementing individual's reasons for implementation.	
Capability	The individual(s) has interpersonal competence, knowledge, and skills to fulfill Role
High-level Leaders	Individuals with a high level of authority, including key decision-makers, executive leaders, or directors
Domain 5. Implementation Process Activities and strategies used to implement the innovation.	
Engaging	Attract and encourage participation in implementation and/or innovation
Motivating	The individual(s) is committed to fulfilling Role
Planning	Identify roles and responsibilities, outline specific steps and milestones, and define goals and measures for implementation success in advance

Note: Definitions are from the updated CFIR published article

APPENDIX 2. Boolean Search Terms

Database	Boolean Search Term
PubMed	((("remote patient monitoring"[MeSH Terms] OR ("remote"[All Fields] AND "patient"[All Fields] AND "monitoring"[All Fields]) OR "remote patient monitoring"[All Fields]) AND ("implementability"[All Fields] OR "implementable"[All Fields] OR "implementation"[All Fields] OR "implementations"[All Fields] OR "implementational"[All Fields] OR "implementations"[All Fields] OR "implementer"[All Fields] OR "implementers"[All Fields] OR "implemment"[All Fields]) AND ("maternal health"[MeSH Terms] OR ("maternal"[All Fields] AND "health"[All Fields]) OR "maternal health"[All Fields])) OR (("telehealth s"[All Fields] OR "telemedicine"[MeSH Terms] OR "telemedicine"[All Fields] OR "telehealth"[All Fields]) AND ("implementability"[All Fields] OR "implementable"[All Fields] OR "implementation"[All Fields] OR "implementations"[All Fields] OR "implementational"[All Fields] OR "implementations"[All Fields] OR "implementer"[All Fields] OR "implementers"[All Fields] OR "implemment"[All Fields]) AND ("obstetric"[All Fields] OR "obstetrically"[All Fields] OR "obstetrics"[MeSH Terms] OR "obstetrics"[All Fields] OR "obstetrical"[All Fields])) OR ("remote patient monitoring"[MeSH Terms] OR ("remote"[All Fields] AND

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	"patient"[All Fields] AND "monitoring"[All Fields]) OR "remote patient monitoring"[All Fields]) AND ("implementability"[All Fields] OR "implementable"[All Fields] OR "implementation"[All Fields] OR "implementations"[All Fields] OR "implementer"[All Fields] OR "implementers"[All Fields] OR "implemention"[All Fields]) AND ("pregnancy"[MeSH Terms] OR "pregnancy"[All Fields] OR "pregnancies"[All Fields] OR "pregnancy s"[All Fields])) AND ((ffrft[Filter]) AND (fft[Filter]) AND (2010/1:2025/12[pdat]))
Science Direct	((remote patient monitoring'/exp OR 'telehealth) AND ('implementation'/exp OR 'provision') AND ('maternal health'/exp OR 'obstetrics')) AND [2010-2025]/py
Embase	('remote patient monitoring'/exp OR 'remote patient monitoring' OR (remote AND ('patient'/exp OR patient) AND ('monitoring'/exp OR monitoring))) AND ('implementation'/exp OR implementation) AND ('maternal health'/exp OR 'maternal health' OR (('maternal'/exp OR maternal) AND ('health'/exp OR health))) AND [2010-2025]/py
Web of Science	((remote patient monitoring'/exp OR 'telehealth) AND ('implementation'/exp OR 'provision') AND ('maternal health'/exp OR 'obstetrics')) AND [2010-2025]/py

413

414 APPENDIX 3. Quality assessment for included quantitative and qualitative, and mixed-methods

415 studies using the MMAT checklist

Qualitative	Are there clear research questions ?	Do the collected data allow to address the research questions ?	Does the qualitative approach answer the research question?	Are the qualitative data collection methods able to address the research question?	Are the findings adequately derived from the data?	Is the interpretation of results sufficiently substantiated by data?	Is there coherence between qualitative data sources & interpretation ?	Quality Score
Dullabh et. al. (2023)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	100%
James et. al. (2025)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	100%
Lawrence et. al. (2023)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	100%
MacEwan et. al. (2024)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	100%
Rajpurkar et. al. (2025)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	100%
Sterba et. al. (2023)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	100%
Quantitative	Are there clear research questions ?	Do the collected data address the research questions ?	Is the sampling strategy relevant to the research question?	Is the sample representative of the target population?	Are the measurements appropriate?	Is the risk of nonresponse bias low?	Is statistical analysis appropriate to answer research question?	Quality Score

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Forna et. al. (2024)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	100%
Klein et. al. (2024)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	100%
Mixed Methods	Are there clear research questions ?	Do the collected data allow to address the research questions ?	Is there adequate rationale for using a mixed methods design to address the research question?	Are the different components of the study effectively integrated to answer the research question?	Are the outputs of the integration of qualitative and quantitative components adequately interpreted?	Are divergences and inconsistencies between quantitative and qualitative results adequately addressed?	Do the different components of the study adhere to the quality criteria of each tradition of the methods involved?	Quality Score
Delgado et. al. (2023)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	100%
Hacker et. al. (2022)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	100%
Hofmann et. al. (2022)	Yes	Yes	Yes	Yes	Yes	No	Yes	86%
Howard et. al. (2025)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	100%
Janssen et. al. (2021)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	100%
Jeganathan et. al. (2020)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	100%
Reisinger-Kindle et. al. (2021)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	100%
Rhoads et. al. (2017)	Yes	Yes	Yes	Yes	No	Yes	Yes	86%
%	100%	100%	100%	100%	94%	94%	100%	98%

416 **Note:** The MMAT checklist uses a scale of Yes, No, or Unsure. References are from the main manuscript.

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418 APPENDIX 4. Quality assessment for included expert opinions using the JBI Critical Appraisal

419 Checklist

	Aziz et. al. (2020)	Peahl et. al. (2020)	Ukoha et. al. (2021)	%
Is the source of the opinion clearly identified?	Yes	Yes	Yes	100%
Does the source of the opinion have standing in the field of expertise?	Yes	Yes	Yes	100%
Are the interests of the relevant population the central focus of the opinion?	Yes	Yes	Yes	100%

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Does the opinion demonstrate a logically defended argument to support the conclusions drawn?	Yes	Yes	Yes	100%
Is there reference to the extant literature?	Yes	Yes	Yes	100%
Is any incongruence with the literature/sources logically defended?	Yes	Yes	Yes	100%
Quality Score	100%	100%	100%	100%

Note: The JBI checklist uses a scale of Yes, No, Unclear, or Not Applicable. References are from the main manuscript.

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Journal Submission:

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- Potential journals for submission:
- **Implementation Science:** \$3490.00 USD
 - **Implementation Science Communications:** \$2390.00 USD
 - **BMC Health Services Research:** \$3450.00 USD
 - **The Journal of mHealth:** \$1650

155 Journal I want to submit to: BMC Health Services Research

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Aims and Scope: *BMC Health Services Research* welcomes articles in the following broad areas of research:

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- Designing and implementing health system reform
- eHealth, mHealth and informatics
- Health policy, politics and power

- 162 • Health system quality and safety
- 163 • Healthcare delivery and access to healthcare
- 164 • Healthcare financing and economics
- 165 • Healthcare governance and management
- 166 • The health workforce

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168 **Formatting:** For the most part, this paper has the necessary sections for BMC. The discussion
169 section can omit “strengths” and only include “limitations.” However, do I need to register this
170 study into PROSPERO? I used PRISMA already, but not sure if that’s enough.